NON-ENDOGENOUS FIXED POINTS OF THE DISCOUNTED TREE SUM EQUATION: THE REGULAR CASE

YITZCHAK SHMALO

ABSTRACT. Aldous and Bandyopadhyay showed that every nonzero solution Y of the homogeneous equation $Y \stackrel{d}{=} \max_i \xi_i Y_i$ produces, for each $a \geq 0$, an invariant measure μ_a of the discounted tree sum equation $X \stackrel{d}{=} \eta + \max_i \xi_i X_i$, and conjectured that for $a > 0$ the invariant recursive tree process with marginal μ_a is never endogenous. We show that the conditional essential supremum of Y given all the innovations is an endogenous solution of the homogeneous equation with values in $[0, \infty]$, and that μ_a fails to be endogenous as soon as this supremum is infinite with positive probability. It follows that the conjecture, for all nonzero Y at once, is equivalent to the absence of a nonnull finite endogenous solution of the homogeneous equation. When the weights have a characteristic exponent, the disintegration theorem of Alsmeyer, Biggins and Meiners shows that the conditional supremum is infinite on the survival event of an endogenous fixed point of the smoothing transform, and the conjecture holds.

1. INTRODUCTION

Let $(\eta, (\xi_i)_{i\geq 1})$ be a random element of $[0, \infty) \times [0, \infty)^{\mathbb{N}}$ and consider the recursive distributional equation

$$X \stackrel{d}{=} \eta + \sup_{i\geq 1} \xi_i X_i \qquad \text{on } [0, \infty), \tag{1}$$

where $X_1, X_2, \ldots$ are independent copies of X, independent of $(\eta, (\xi_i))$. Equation (1) is the discounted tree sum equation of Aldous and Bandyopadhyay [1, Section 4.4]. On the tree of finite words over $\mathbb{N}$, place independent copies $(\eta_v, (\xi_{vi})_{i\geq 1})$ of the innovation at the vertices v, and let $L(v)$ be the product of the weights along the path from the root $\varnothing$ to v, with $L(\varnothing) = 1$. The variable

$$X = \sup_{\varnothing = v_0, v_1, v_2, \ldots} \sum_{d\geq 0} \eta_{v_d} L(v_d), \tag{2}$$

the supremum being over rays from the root, is a solution of (1) whenever it is finite almost surely, and its law μ is the lower invariant measure. When the weights come from a branching random walk, the finiteness of X is studied by Aïdékon, Hu and Shi [3], who partly answer Open Problem 31 of [1]. Dropping η gives the homogeneous equation

$$Y \stackrel{d}{=} \sup_{i\geq 1} \xi_i Y_i \qquad \text{on } [0, \infty). \tag{3}$$

Aldous and Bandyopadhyay proved the following [1, Corollary 33]: if X in (2) is finite almost surely and $\operatorname{dist}(Y)$ is a nonzero solution of (3), then for every $a \geq 0$ the iterates of the map of (1) started at $\operatorname{dist}(aY)$ converge to an invariant measure μ_a; if moreover

$$\eta \text{ is independent of } (\xi_i), \text{ and } 0 \text{ is in the support of } \eta, \tag{4}$$

then the measures μ_a are distinct. They stated the following.

Conjecture 1.1 ([1, Conjecture 34])**.** *Under the assumptions of Corollary 33 of* [1] *and* (4)*, for $a > 0$ the invariant recursive tree process associated with μ_a is not endogenous.*

Endogeny means that the value at the root is a measurable function of the innovations [1, Definition 7]; the precise setting is recalled in Section 2. Mach, Sturm and Swart [7] characterize endogeny through a higher-level equation without the continuity assumption of [1], and Ráth, Swart and Szőke [9] give a recursive tree process that passes from endogeny to nonendogeny as a parameter varies. Let $\mathcal{G}$ be the σ-field generated by all the innovations $(\eta_v, (\xi_{vi})_i)$, and let (Y_v) be the invariant recursive tree process of (3) with marginal $\mathrm{dist}(Y)$, built on the same tree. Put

$$M = \operatorname{ess\,sup}(Y_\varnothing \mid \mathcal{G}) \in [0, \infty].$$

Our first result reduces the conjecture to a statement about (3) alone.

Theorem 1.2. *Suppose X in (2) is finite almost surely and let $\mathrm{dist}(Y)$ be a nonzero solution of (3).*

(a) *M is an endogenous solution of (3) on $[0,\infty]$, and $M \geq Y_\varnothing$ almost surely.*
(b) *If the recursive tree process of Y is endogenous, then μ_a is endogenous for every $a \geq 0$.*
(c) *If $\mathbb{P}(M = \infty) > 0$, then μ_a is not endogenous for any $a > 0$.*

Consequently the following are equivalent:

(i) *for every nonzero solution $\mathrm{dist}(Y)$ of (3) and every $a > 0$, the measure μ_a is not endogenous;*
(ii) *the only endogenous solution of (3) on $[0,\infty)$ is δ_0.*

Condition (4) plays no role in Theorem 1.2. Our second result settles Conjecture 1.1 when the weights have a characteristic exponent. Let $N = \#\{i : \xi_i > 0\}$ and

$$m(\theta) = \mathbb{E} \sum_{i:\, \xi_i > 0} \xi_i^\theta \in [0, \infty], \qquad \theta \geq 0.$$

Theorem 1.3. *Suppose X in (2) is finite almost surely, that $1 < m(\theta_0) < \infty$ for some $\theta_0 \geq 0$ and $m(p) < 1$ for some $p > \theta_0$, and that either $N < \infty$ almost surely or $\mathbb{E}\, W_1 \log^+ W_1 < \infty$, where $W_1 = \sum_i \xi_i^\alpha$ and α is the unique point of (θ_0, p) with $m(\alpha) = 1$. Then for every nonzero solution $\mathrm{dist}(Y)$ of (3) we have $\mathbb{P}(M = \infty) > 0$, and for every $a > 0$ the invariant recursive tree process associated with μ_a is not endogenous.*

The proof identifies M through the disintegration of fixed points of the smoothing transform due to Alsmeyer, Biggins and Meiners [2, Theorem 8.3]: conditionally on the weights, $1/Y_\varnothing$ has a Weibull-type survival function $\exp(-Wh(t)t^\alpha)$ with W an endogenous fixed point of the smoothing transform with weights (ξ_i^α), so $Y_\varnothing$ is conditionally unbounded on $\{W > 0\}$. Theorem 1.3 covers Examples 37 and 38 of [1]; see Section 5, which also discusses what remains open.

2. Preliminaries

2.1. **Recursive tree processes.** Let $V = \bigcup_{n \geq 0} \mathbb{N}^n$ be the set of finite words, with root $\varnothing$, children vi of v, and $|v|$ the length of v. Let $\theta_v = (\eta_v, (\xi_{vi})_{i \geq 1})$, $v \in V$, be independent copies of $(\eta, (\xi_i))$. For $v \in V$ let $\mathcal{G}_v$ be the σ-field generated by $(\theta_w : w \in vV)$, the innovations of the subtree rooted at v, so that $\mathcal{G} = \mathcal{G}_\varnothing$. For $w \in vV$ write $L_v(w)$ for the product of the weights along the path from v to w, with $L_v(v) = 1$, and $L = L_\varnothing$.

Consider a recursive distributional equation of the form $Z \stackrel{d}{=} g(\theta; Z_1, Z_2, \dots)$. An invariant recursive tree process with marginal ν is a family $(Z_v)_{v \in V}$, defined jointly with (θ_v), such that $Z_v = g(\theta_v; Z_{v1}, Z_{v2}, \dots)$ almost surely for every v, and such that for every n the variables $(Z_v : |v| = n)$ are independent with law ν and independent of $(\theta_w : |w| < n)$. By [1, Lemma 6] such a process exists if and only if ν is a solution. Its joint law with the innovations is determined by ν: it is determined by the laws of the families $(\theta_w : |w| < k;\ Z_w : |w| = k)$, $k \geq 0$, which are product laws, since every Z_w with $|w| < k$ is almost surely a function of such a family. The process is endogenous if $Z_\varnothing$ is $\mathcal{G}$-measurable (up to a null set); by the uniqueness just noted this is a property of ν, and we say that ν is endogenous.

Lemma 2.1. *Let (Z_v) be an invariant recursive tree process and, for $v \in V$, let $\mathcal{F}_v$ be the σ-field generated by $(\theta_w, Z_w : w \in vV)$. For every n, the σ-fields $\mathcal{F}_v$, $|v| = n$, and $\sigma(\theta_w : |w| < n)$ are independent, and the family $(\theta_{vw}, Z_{vw})_{w\in V}$ has the same law as $(\theta_w, Z_w)_{w\in V}$ for every v.*

Proof. For $k \geq n$, the family $(\theta_w : |w| < k;\ Z_w : |w| = k)$ consists of independent variables. Grouping it according to the vertex of depth n above each w with $|w| \geq n$ gives independent blocks, independent of $(\theta_w : |w| < n)$, and the block of v has the law of the family $(\theta_w : |w| < k - n;\ Z_w : |w| = k - n)$. The variables Z_w with $n \leq |w| \leq k$ are almost surely functions of these blocks. As $k \to \infty$ the σ-fields generated by the blocks increase and generate $\mathcal{F}_v$ up to null sets, and the claims follow. □

We use the process of (3) with innovations θ_v; the map of (3) does not use η_v.

Lemma 2.2. *Let (Y_v) be the invariant recursive tree process of (3) with innovations (θ_v). Conditionally on $\sigma(\xi_{vi} : v \in V, i \geq 1)$, the families (Y_v) and (η_v) are independent. In particular $\mathbb{P}(Y_\varnothing > y \mid \mathcal{G}) = \mathbb{P}(Y_\varnothing > y \mid \xi_{vi} : v \in V, i \geq 1)$ almost surely for every y.*

Proof. Take an invariant recursive tree process (Y_v) of (3) with innovations $(\xi_{vi})_i$ only, and draw each η_v from the conditional law of η given $(\xi_i) = (\xi_{vi})_i$, independently over v and of (Y_v) given the weights. The pairs θ_v are then independent copies of $(\eta, (\xi_i))$, and for every n the variables $(Y_v : |v| = n)$ are independent of $(\theta_w : |w| < n)$, since $(\eta_w : |w| < n)$ is a function of $(\xi_{wi} : |w| < n)$ and of randomness independent of everything else. So this is an invariant recursive tree process with innovations (θ_v), it has the stated conditional independence, and by uniqueness in law every such process has it. □

2.2. **Conditional essential supremum.** For a $[0,\infty]$-valued random variable U and a σ-field $\mathcal{H}$, the conditional essential supremum $\operatorname{ess\,sup}(U \mid \mathcal{H})$ is the $\mathcal{H}$-measurable $[0,\infty]$-valued variable S, unique up to null sets, such that $U \leq S$ almost surely and $S \leq S'$ almost surely for every $\mathcal{H}$-measurable S' with $U \leq S'$ almost surely. One may take $S = \sup\{q \in \mathbb{Q}_+ : \mathbb{P}(U > q \mid \mathcal{H}) > 0\}$ with $\sup \varnothing = 0$. We use three elementary facts, valid almost surely:

(E1) $\operatorname{ess\,sup}(\sup_i U_i \mid \mathcal{H}) = \sup_i \operatorname{ess\,sup}(U_i \mid \mathcal{H})$ for countably many $U_i \geq 0$;
(E2) $\operatorname{ess\,sup}(cU \mid \mathcal{H}) = c \operatorname{ess\,sup}(U \mid \mathcal{H})$ for $\mathcal{H}$-measurable $c \in [0,\infty)$, with $0 \cdot \infty = 0$;
(E3) if $\sigma(U) \vee \mathcal{H}_1$ is independent of $\mathcal{H}_2$, then $\operatorname{ess\,sup}(U \mid \mathcal{H}_1 \vee \mathcal{H}_2) = \operatorname{ess\,sup}(U \mid \mathcal{H}_1)$.

(E1) and (E2) follow from the definition, and (E3) from $\mathbb{P}(U > q \mid \mathcal{H}_1 \vee \mathcal{H}_2) = \mathbb{P}(U > q \mid \mathcal{H}_1)$.

3. The conditional supremum and the coupling

Throughout this section X in (2) is finite almost surely, $\operatorname{dist}(Y)$ is a nonzero solution of (3), and (Y_v) is its invariant recursive tree process with innovations (θ_v). Fix a measurable m_0 with $\operatorname{ess\,sup}(Y_\varnothing \mid \mathcal{G}) = m_0((\theta_w)_{w\in V})$ and put $M_v = m_0((\theta_{vw})_{w\in V})$.

Lemma 3.1. *Almost surely, $M_v = \operatorname{ess\,sup}(Y_v \mid \mathcal{G})$ for every v, and $M_v = \sup_i \xi_{vi} M_{vi}$. The family (M_v) is an endogenous invariant recursive tree process of (3) on $[0,\infty]$, and $M_\varnothing \geq Y_\varnothing$.*

Proof. By Lemma 2.1, $\sigma(Y_v) \vee \mathcal{G}_v \subset \mathcal{F}_v$ is independent of the σ-field generated by the innovations outside the subtree of v, and the law of $(Y_v, (\theta_{vw})_w)$ is that of $(Y_\varnothing, (\theta_w)_w)$. By (E3), $\operatorname{ess\,sup}(Y_v \mid \mathcal{G}) = \operatorname{ess\,sup}(Y_v \mid \mathcal{G}_v) = M_v$. Since $Y_v = \sup_i \xi_{vi} Y_{vi}$ almost surely and each ξ_{vi} is $\mathcal{G}$-measurable, (E1) and (E2) give $M_v = \sup_i \xi_{vi} M_{vi}$. The variables $(M_v : |v| = n)$ are functions of the innovations of disjoint subtrees of depth-n vertices, so they are independent, identically distributed and independent of $(\theta_w : |w| < n)$. Thus (M_v) is an invariant recursive tree process of (3) on $[0,\infty]$, and it is $\mathcal{G}$-measurable. Finally $Y_\varnothing \leq M_\varnothing$ by definition. □

For $a \geq 0$, $n \geq 0$ and $v \in V$ let

$$Z_{a,v}(n) = \sup_{v=w_0,w_1,\ldots,w_n} \Big(aY_{w_n} L_v(w_n) + \sum_{d=0}^{n-1} \eta_{w_d} L_v(w_d) \Big),$$

the supremum being over paths of length n starting at v, each w_j a child of w_{j-1}. For $v = \varnothing$ this is the variable $Z_a(n)$ of the proof of [1, Corollary 33], whose law is the n-th iterate of the map of (1) at $\mathrm{dist}(aY)$. Let X_v be the variable (2) computed in the subtree of v.

Lemma 3.2. *Almost surely, for every v the limit $Z_{a,v} = \lim_n Z_{a,v}(n)$ exists, and*

$$aY_v \leq Z_{a,v} \leq X_v + aY_v < \infty. \tag{5}$$

Moreover $Z_{a,v} = \eta_v + \sup_i \xi_{vi} Z_{a,vi}$, and $(Z_{a,v})$ is an invariant recursive tree process of (1) with marginal μ_a.

Proof. Iterating $Y_w = \sup_i \xi_{wi} Y_{wi}$ gives $aY_v = \sup_{|w|=|v|+n,\, w \in vV} aY_w L_v(w)$, and since $\eta \geq 0$ this is at most $Z_{a,v}(n)$. Bounding the supremum of a sum by the sum of the suprema gives $Z_{a,v}(n) \leq aY_v + X_v$, because each partial sum along a path is at most the full sum along any ray extending it. Aldous and Bandyopadhyay show that $Z_{a,v}(n)$ is nondecreasing in n: given w_n, a child w_{n+1} with $Y_{w_n} = \xi_{w_{n+1}} Y_{w_{n+1}}$ up to an arbitrarily small error extends the path without decreasing the bracket. This gives the limit and (5). Splitting off the first step of each path,

$$Z_{a,v}(n) = \eta_v + \sup_i \xi_{vi} Z_{a,vi}(n-1),$$

and since the terms increase in n, the suprema and the limit commute, which gives the recursion. Each $Z_{a,v}$ is the same measurable function of $(\theta_{vw}, Y_{vw})_{w \in V}$, so by Lemma 2.1 the variables $(Z_{a,v} : |v| = n)$ are independent with a common law, independent of $(\theta_w : |w| < n)$. The common law is $\lim_n \mathrm{dist}(Z_a(n)) = \mu_a$. □

Proof of Theorem 1.2. Part (a) is Lemma 3.1.

(b) If (Y_v) is endogenous then each Y_v is $\mathcal{G}$-measurable, by Lemma 2.1 applied to the shifted process. Then $Z_{a,\varnothing}(n)$, a supremum of countably many $\mathcal{G}$-measurable variables, is $\mathcal{G}$-measurable, and so is $Z_{a,\varnothing}$. By Lemma 3.2 and uniqueness in law, μ_a is endogenous.

(c) Let $a > 0$ and suppose μ_a is endogenous. The process $(Z_{a,v})$ of Lemma 3.2 has marginal μ_a, so $Z_{a,\varnothing} = f(\mathcal{G})$ almost surely for a measurable f, and $f(\mathcal{G}) < \infty$ by (5). Then $Y_\varnothing \leq f(\mathcal{G})/a$ almost surely, so $M_\varnothing \leq f(\mathcal{G})/a < \infty$ almost surely, contradicting $\mathbb{P}(M = \infty) > 0$.

For the equivalence, suppose (ii) fails and let $\mathrm{dist}(Y^*) \neq \delta_0$ be an endogenous solution of (3) on $[0, \infty)$. Taking $Y = Y^*$ in (b), μ_a is endogenous for every a, so (i) fails. Conversely, suppose (ii) holds and let $\mathrm{dist}(Y)$ be any nonzero solution. By (a), M is an endogenous solution on $[0, \infty]$ with $M \geq Y_\varnothing$, so $\mathrm{dist}(M) \neq \delta_0$. If M were finite almost surely it would be an endogenous solution on $[0, \infty)$ other than δ_0; hence $\mathbb{P}(M = \infty) > 0$, and (c) gives (i). □

4. The regular case

We use the setting of [2] with weights $T_i = \xi_i$. In their notation, the assumptions are

(A1) $\mathbb{P}((\xi_i) \in \{0,1\}^{\mathbb{N}}) < 1$;
(A2) $\mathbb{E}N > 1$;
(A3) there is $\alpha > 0$ with $1 = m(\alpha) < m(\beta)$ for all $\beta \in [0, \alpha)$;
(A4a) $m'(\alpha) = \mathbb{E} \sum_i \xi_i^\alpha \log \xi_i \in (-\infty, 0)$ and $\mathbb{E}\, W_1 \log^+ W_1 < \infty$;
(A4b) $m(\theta) < \infty$ for some $\theta \in [0, \alpha)$;
(A5) there is a finite, nonnegative random variable W with $\mathbb{P}(W \neq 0) > 0$ and $W \overset{d}{=} \sum_i \xi_i^\alpha W_i$.

Lemma 4.1. *Under the hypotheses of Theorem 1.3, the point α is well defined and (A1)–(A3), (A4b) and (A5) hold.*

Proof. The function m is convex from $[0,\infty)$ to $[0,\infty]$, finite on $[\theta_0,p]$, with $m(\theta_0)>1>m(p)$. Hence there is $\alpha\in(\theta_0,p)$ with $m(\alpha)=1$. If $\beta\in(\theta_0,\alpha)$ had $m(\beta)\le 1$, convexity on $\beta<\alpha<p$ would give $m(\alpha)<1$; if $\beta\in[0,\theta_0)$ had $m(\beta)\le 1$, convexity on $\beta<\theta_0<\alpha$ would give $m(\theta_0)\le 1$. So $m>1$ on $[0,\alpha)$, which is (A3) and shows that α is unique; in particular $\mathbb{E}N=m(0)>1$, which is (A2), and (A4b) holds with $\theta=\theta_0$. If all weights lay in $\{0,1\}$, then m would be constant, which it is not; this is (A1). For θ in a compact subinterval of (θ_0,p), $\xi^\theta|\log\xi|$ is bounded by a constant times $\xi^{\theta_0}+\xi^p$, so by dominated convergence m is differentiable at α with $m'(\alpha)=\mathbb{E}\sum_i \xi_i^\alpha\log\xi_i$, and by convexity $(m(\alpha)-m(\theta_0))/(\alpha-\theta_0)\le m'(\alpha)\le(m(p)-m(\alpha))/(p-\alpha)<0$. So the alternative $\mathbb{E}W_1\log^+W_1<\infty$ of the hypotheses gives (A4a). Now (A5) follows from [2, Proposition 2.1], which requires (A1)–(A3) together with either (A4a) or $N<\infty$ almost surely. □

Lemma 4.2. *If $m(p)<1$ for some $p>0$, every nonzero solution of (3) has unbounded support; in particular it is not concentrated on a single point.*

Proof. If $Y\le c$ almost surely, then $Y_\varnothing=\sup_{|v|=n}L(v)Y_v\le c\sup_{|v|=n}L(v)$, and $\mathbb{E}\sup_{|v|=n}L(v)^p\le\mathbb{E}\sum_{|v|=n}L(v)^p=m(p)^n\to 0$. Hence $Y_\varnothing=0$ almost surely. □

Proof of Theorem 1.3. Let $\mathrm{dist}(Y)$ be a nonzero solution of (3) with process (Y_v), and put $Y'_v=1/Y_v\in(0,\infty]$. Then $Y'_v=\inf\{Y'_{vi}/\xi_{vi}:\xi_{vi}>0\}$ with $\inf\varnothing=\infty$, which is equation (1.3) of [2]. Let $f(t)=\mathbb{P}(Y'_\varnothing\ge t)$ for $t\ge 0$. Then f is nonincreasing with $f(0)=f(0+)=1$, and $f(t)=\mathbb{E}\prod_i f(t\xi_i)$ with $f(0)=1$. If f took only the values 0 and 1 on $(0,\infty)$, then $Y'_\varnothing$ would be almost surely constant in $(0,\infty]$, so Y would be almost surely 0 or almost surely a positive constant, which is excluded by the hypothesis and by Lemma 4.2. So f belongs to the class $\mathcal{S}(\mathcal{M})$ of [2].

Let $\mathcal{A}_n$ be the σ-field generated by $(\xi_{vi}:|v|<n,\ i\ge 1)$ and $\mathcal{A}_\infty=\bigvee_n\mathcal{A}_n$. Iterating the equation, $Y'_\varnothing=\inf\{Y'_v/L(v):|v|=n,\ L(v)>0\}$, and the variables $(Y'_v:|v|=n)$ are independent with survival function f and independent of $\mathcal{A}_n$. Hence

$$\mathbb{P}(Y'_\varnothing\ge t\mid\mathcal{A}_n)=\prod_{|v|=n}f(tL(v))=:M_n(t),$$

the multiplicative martingale of [2, Section 8]. By Lévy's upward theorem, $M_n(t)\to\mathbb{P}(Y'_\varnothing\ge t\mid\mathcal{A}_\infty)$ almost surely. By Lemma 4.1 and [2, Theorem 8.3], there are an endogenous fixed point W as in (A5) and a function h in the class $\mathcal{H}_r$ of [2] (multiplicatively r-periodic, with $h(t)t^\alpha$ nondecreasing) such that, for every $t\ge 0$,

$$\mathbb{P}(Y'_\varnothing\ge t\mid\mathcal{A}_\infty)=e^{-Wh(t)t^\alpha}\quad\text{almost surely.}$$

The function h is positive: if $h(t_0)=0$, then periodicity and monotonicity of $h(t)t^\alpha$ force $h\equiv 0$, and taking expectations would give $f\equiv 1$. By Lemma 2.2, conditioning on $\mathcal{G}$ is the same as conditioning on $\mathcal{A}_\infty$, so for every rational $y>0$,

$$\mathbb{P}(Y_\varnothing>y\mid\mathcal{G})=1-e^{-Wh(1/y)y^{-\alpha}}>0\qquad\text{almost surely on }\{W>0\}.$$

Hence $M=\infty$ almost surely on $\{W>0\}$, an event of positive probability, and Theorem 1.2(c) completes the proof. □

5. Examples and remarks

Example 5.1. Let every vertex have four children with weight $1/2$. Then $m(\theta)=4\cdot 2^{-\theta}$ and $\alpha=2$. The Fréchet law $\mathbb{P}(Y\le y)=e^{-y^{-2}}$ solves (3), and (Y_v) is independent of $\mathcal{G}$, so $M=\infty$ and, for any bounded $\eta\ge 0$, no μ_a with $a>0$ is endogenous. In contrast, with two children of weight 1 and $\eta=0$ the constant $Y=1$ is an endogenous solution and $\mu_a=\delta_a$ is endogenous for every a, as Theorem 1.2(b) predicts; here $m\equiv 2$ and Theorem 1.3 does not apply.

Example 5.2. In Example 37 of [1], $\xi_i = c \in (0,1)$ for $i \le N$, where N is the offspring variable of a Galton–Watson tree. If $1 < \mathbb{E}N < \infty$ then $m(\theta) = c^\theta \mathbb{E}N$, so Theorem 1.3 applies with $\theta_0 = 0$ and $\alpha = \log \mathbb{E}N / \log(1/c)$. In Example 38 of [1], $\xi_i = e^{-\zeta_i}$ with (ζ_i) the points of a rate one Poisson process; then $N = \infty$, $m(\theta) = 1/\theta$, $\alpha = 1$, and $W_1 = \sum_i e^{-\zeta_i}$ has finite second moment, so the hypotheses hold. In both examples every invariant measure μ_a, $a > 0$, built from a nonzero solution of (3) is non-endogenous.

Remark 5.3. In the regular case the dichotomy in Theorem 1.2 does not leave room for an intermediate situation: the process of Y is never endogenous and M is infinite with positive probability. Outside the regular case, a counterexample to Conjecture 1.1 must come from an endogenous solution of (3) on $[0,\infty)$ other than δ_0, and when $m(p) < 1$ for some $p > 0$ such a solution must have unbounded support by Lemma 4.2. Two cases remain open: weights without a characteristic exponent, where m jumps from ∞ to a value at most 1 (the case studied for the min-type equation by Jagers and Rösler [5] and Alsmeyer and Meiners [4]), and infinitely many positive weights for which $\mathbb{E}W_1 \log^+ W_1 = \infty$.

Remark 5.4. The correspondence between solutions of an inhomogeneous max-type equation and its homogeneous counterpart was studied by Rüschendorf [8], and the tails of the minimal solution of $R \overset{d}{=} Q \vee \bigvee_i C_i R_i$ were determined by Jelenković and Olvera-Cravioto [6]. Theorem 1.2 transfers endogeny along the correspondence of [1, Corollary 33] in one direction and transfers its failure through the conditional supremum M in the other.

Acknowledgments and funding

This research was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [10] and working together inside the Brain framework [12], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [11], the Math Brain repository [10] and the Brain framework repository [12].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

Anthropic Claude AI systems were used to develop the mathematical argument, check the cited results, and write and prepare this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for publication rests with the author. The AI-assisted internal reviews and compilation checks are not external peer review or formal verification.

Data and code availability

No data or code were used. The source of this manuscript is in the repository

`github.com/yspennstate/discounted-tree-sums-endogeny`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [10].

References

[1] D. J. Aldous and A. Bandyopadhyay, A survey of max-type recursive distributional equations, *Ann. Appl. Probab.* 15 (2005), 1047–1110.

[2] G. Alsmeyer, J. D. Biggins and M. Meiners, The functional equation of the smoothing transform, *Ann. Probab.* 40 (2012), 2069–2105.

[3] E. Aïdékon, Y. Hu and Z. Shi, Boundedness of discounted tree sums, *Acta Math. Sin. (Engl. Ser.)* 42 (2026), 1639–1661. arXiv:2409.01048.

[4] G. Alsmeyer and M. Meiners, A min-type stochastic fixed-point equation related to the smoothing transformation, *Theory Stoch. Process.* 15 (2009), 19–41.

[5] P. Jagers and U. Rösler, Stochastic fixed points for the maximum, in *Mathematics and Computer Science III*, Trends Math., Birkhäuser, Basel, 2004, 325–338.

[6] P. R. Jelenković and M. Olvera-Cravioto, Maximums on trees, *Stochastic Process. Appl.* 125 (2015), 217–232.

[7] T. Mach, A. Sturm and J. M. Swart, A new characterization of endogeny, *Math. Phys. Anal. Geom.* 21 (2018), Paper No. 30. arXiv:1801.05253.

[8] L. Rüschendorf, On stochastic recursive equations of sum and max type, *J. Appl. Probab.* 43 (2006), 687–703.

[9] B. Ráth, J. M. Swart and M. Szőke, A phase transition between endogeny and nonendogeny, *Electron. J. Probab.* 27 (2022), 1–43. arXiv:2103.14408.

[10] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[11] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[12] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.